ME 656: Autonomous Navigation for Mobile Robots

Simulation Lab 1 Report

Course:	ME 656 - Autonomous Navigation for Mobile Robots
Semester:	Spring 2026
Assignment:	SimLab 1
Student:	Matthew Lepis

1 Overview

This report summarizes the five requested deliverables for SimLab 1. All simulations were run with the assignment parameters: a 10 m hallway, a sensing range of 0.5 m, landmarks at 2 m, 5 m, and 8 m, a time step of 0.1 s, and a robot speed of 0.1 m/s. For every simulation, the estimator was evaluated over $N = 1000$ Monte Carlo trials and compared against the known ground truth trajectory.

The least squares approach solves for the full trajectory in batch form, while the Kalman filter approach estimates the state incrementally over time. In all cases, the reported error metric is the mean absolute robot position error as a function of time.

2 Deliverable 1: Batch Least Squares with Odometry Only

For the first simulation, the batch least squares problem was built using only the initial pose constraint and the odometry displacement equations between sequential robot position states. In this simulation variant, no landmark observations are included, therefore the estimate is driven solely by noisy odometry measurements.

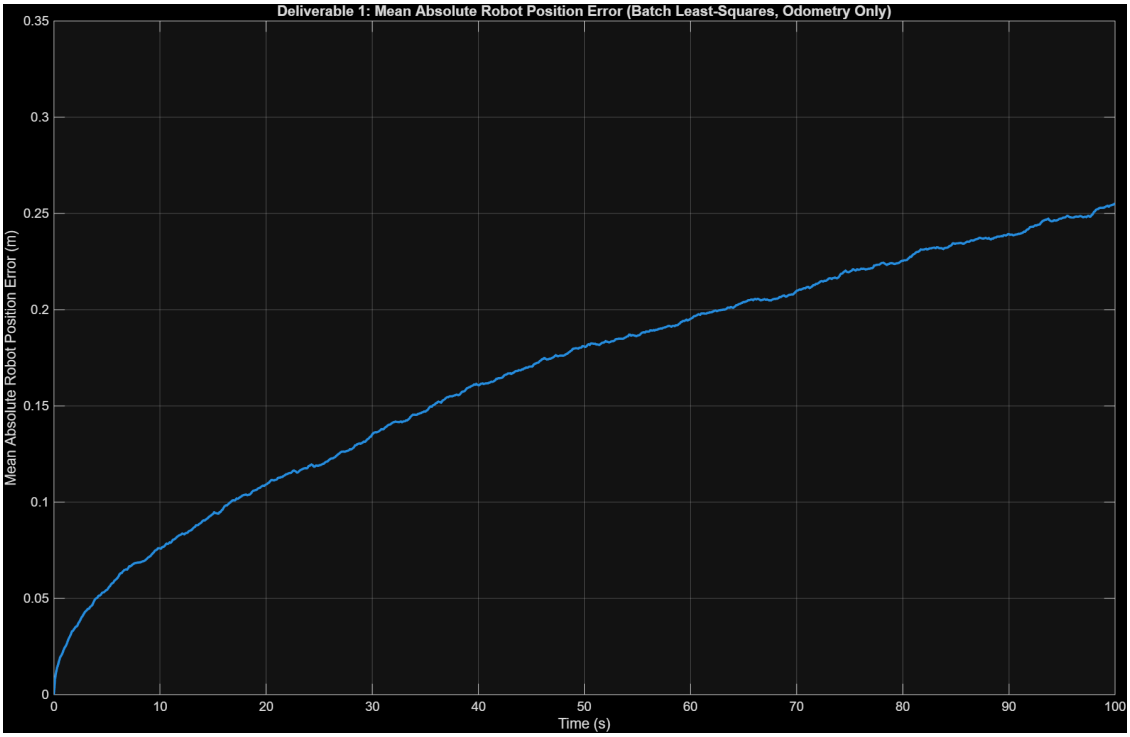

Figure 1: Deliverable 1: Mean absolute robot position error for batch least squares with odometry only.

The mean absolute error grows with time because the robot never receives external positional correction. Each noisy odometry measurement contributes a small amount of drift, and the accumulated error becomes larger as the robot moves farther down the hallway.

3 Deliverable 2: Batch Least Squares with Odometry and Landmark Sensing

In the second simulation, the batch least squares formulation was extended to include signed landmark range measurements of the form $l_j - x_k$ (landmark [j] position minus robot position state [k]). This keeps the measurement model linear while allowing the batch solve to estimate both the robot trajectory and the three landmark positions.

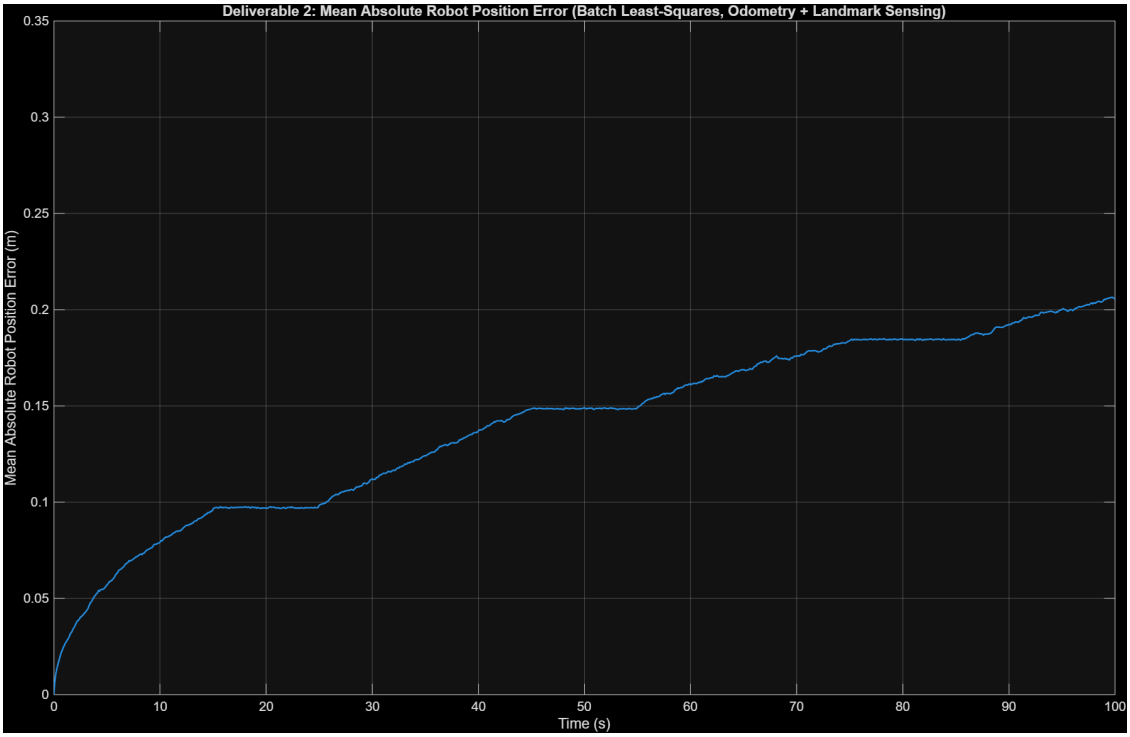

Figure 2: Deliverable 2: Mean absolute robot position error for batch least squares with odometry and landmark sensing.

The introduction of the robot's observation of landmarks reduces the growth of mean error. When landmarks are observed, the batch system gains additional constraints that tie the robot's pose states to fixed features in the hallway. This limits the amount of odometry drift that can accumulate and keeps the mean absolute error lower than in the odometry-only case. Between the odometry-only case and odometry + landmark case, this case had a 19% decrease in peak mean position error.

4 Deliverable 3: Kalman Filter with Odometry Only

In simulation three, a linear discrete-time Kalman filter was implemented with state vector

$$\hat{x}_k = [\hat{\dot{x}}_k \; \hat{x}_k \; \hat{l}_{1,k} \; \hat{l}_{2,k} \; \hat{l}_{3,k}]^T.$$

The filter uses odometry as a direct noisy measurement of the robot velocity state and propagates uncertainty through the covariance matrix P_k. The process noise covariance was chosen according to the assignment, with only $Q(1,1)$ nonzero (10^{-8}).

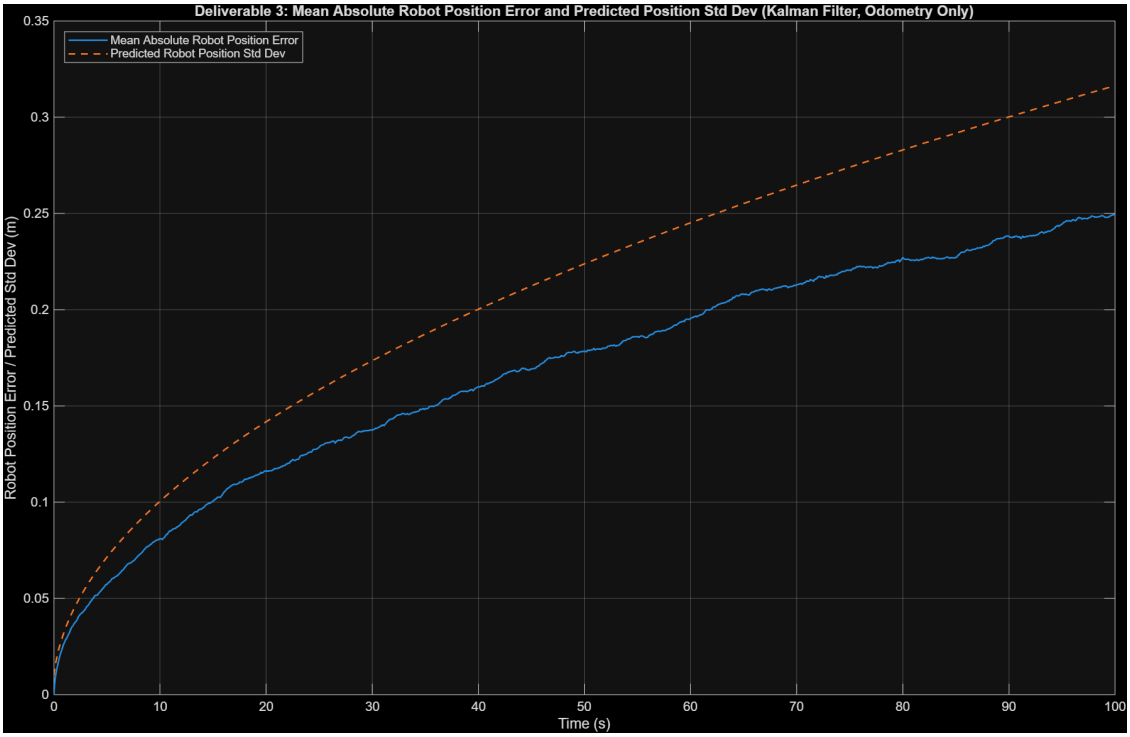

Figure 3: Deliverable 3: Mean absolute robot position error and predicted position standard deviation for the odometry-only Kalman filter.

Compared with the first simulation (Least Squares Error - Odometry Only), the odometry-only Kalman filter shows the same overall behavior: error grows over time because the estimator relies only on noisy motion information and receives no landmark-based correction - nothing to ground the robot back into the space it is traversing. The main additional benefit of the Kalman filter is that it also predicts its own uncertainty through P_k. In the plot, the covariance-based position standard deviation increases with time in a manner that is consistent with the rising empirical error.

5 Deliverable 4: Kalman Filter with Odometry and Landmark Sensing

For simulation four, the Kalman filter was extended to use both odometry and landmark measurements. The measurement matrix H_k and noise covariance R_k were allowed to change dimension with time depending on which landmarks were within sensor range at each time step.

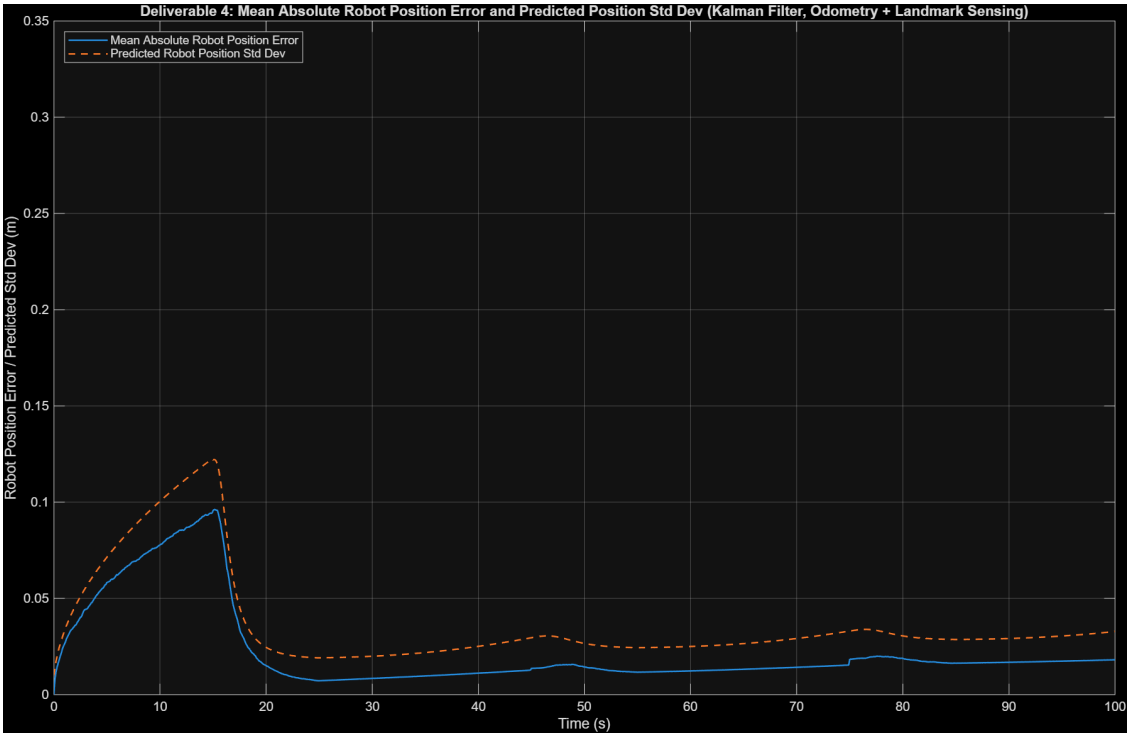

Figure 4: Deliverable 4: Mean absolute robot position error and predicted position standard deviation for the Kalman filter with landmark sensing.

The robot's observation of landmarks clearly reduces the growth of error. Between landmark sightings, the filter behaves similarly to the odometry-only case and uncertainty gradually increases. This odometry-only-like behavior is most clear before it sees a single landmark. When a landmark actually comes into range, the extra measurement constrains the robot position and the predicted standard deviation drops. The empirical mean absolute error also decreases or grows more slowly in those regions.

In the results obtained here, the Kalman filter also outperforms the corresponding batch least squares result. Comparing simulation four with simulation two, the landmark-aware Kalman filter maintains a substantially lower error level over the hallway traversal, making it the strongest one-way estimator in these simulations.

Unlike the Kalman filter, which only uses the information available up to the current time step, the batch least squares formulations with landmark sensing solve for the full trajectory and landmark states simultaneously, so measurements from later times can influence earlier pose estimates through the global solution. In spite of that, the Kalman filter produced the lower error in these runs, while also providing a covariance-based uncertainty estimate that the least squares plots do not directly provide.

6 Deliverable 5: Batch Least Squares with Back-and-Forth Loop Closures

For simulation five, the batch least squares formulation from simulation two was extended so the robot traveled to the end of the hallway and then returned to the origin at -0.1 m/s. This doubles the traversal duration and causes the robot to re-observe the same landmarks on the return trip.

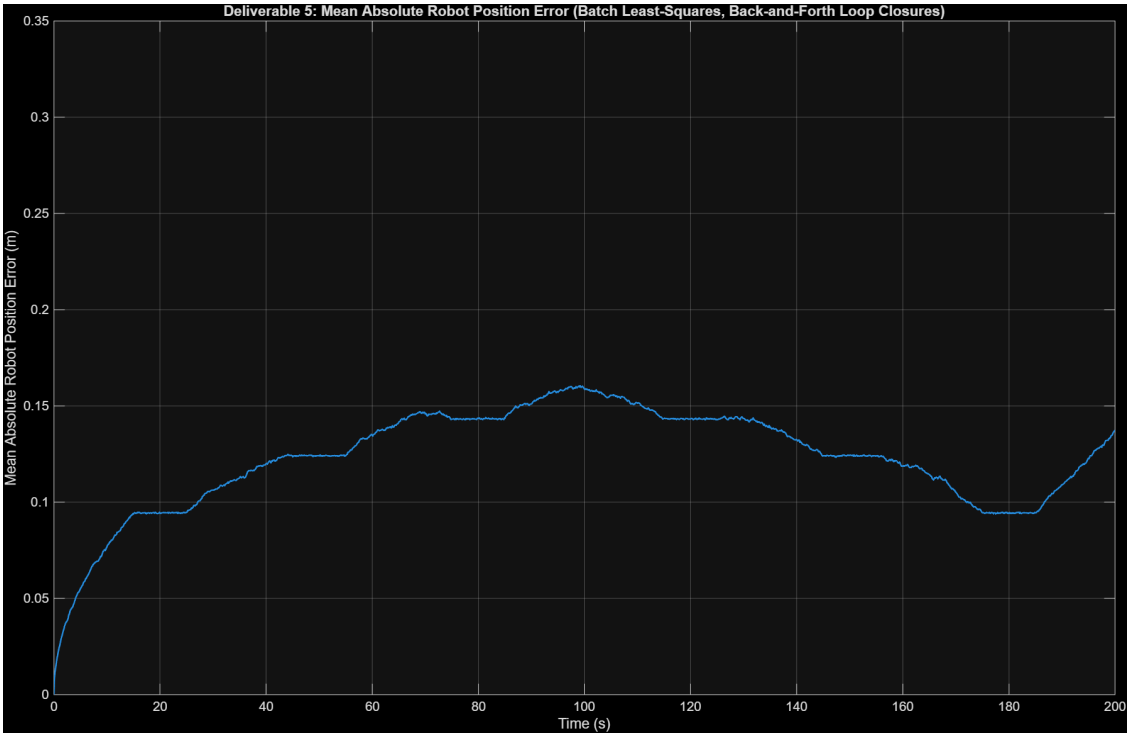

Figure 5: Deliverable 5: Mean absolute robot position error for batch least squares with back-and-forth landmark re-observations.

The re-observation of landmarks does have a meaningful impact on the robot's ability to curb the growth of error. These repeated landmark observations act as loop-closure constraints: later robot poses are connected back to the same fixed landmark states observed earlier in the run. As a result, accumulated drift is reduced on the return trip, and the full back-and-forth trajectory estimate is better constrained than it would be without those repeated observations.

7 Conclusion

Across all five simulations, the main pattern is consistent: odometry alone leads to steadily increasing localization error, while landmark observations reduce drift by anchoring the robot to fixed features in the hallway. In the results obtained during these simulations, the Kalman filter outperformed the batch least squares approach overall, with the landmark-aware Kalman filter in simulation four producing the lowest sustained error among the one-way hallway traversals.

The back-and-forth batch least squares case in simulation five still demonstrates an important SLAM effect: repeated landmark observations on the return trip help reduce accumulated drift through loop-closure-style constraints. Even so, based on the plots generated here, the Kalman filter produced the best overall localization performance in this project while also providing an explicit covariance-based uncertainty estimate.

8 Future Work

If this work were extended beyond the guiding requirements, several next steps would be valuable. First, implementing a return run for the Kalman filter would make it possible to compare a back-

and-forth recursive estimator directly against Deliverable 5 and study whether loop-closure-like repeated observations provide similar benefits in the incremental setting. Second, the batch least squares implementations in simulation one, two, and five could be refactored for readability, since the Kalman filter implementations developed later in the assignment are more modular and easier to interpret at a glance.

Additional simulation studies would also help characterize estimator behavior more deeply. Varying the number and placement of landmarks would show how map feature density affects localization accuracy. Adjusting robot properties such as its speed and the measurement update interval would also be useful, since changing the traversal speed or the time-step would directly affect how quickly error accumulates and how often each estimator receives information. Plotting the estimated state against the true state for each variation would make the trajectory differences more visually interpretable than error-only plots. Finally, extending the run time of each simulation would help evaluate long-horizon error resilience and reveal how each method behaves when drift has more time to accumulate.

9 Included MATLAB Files

The MATLAB source associated with this report is organized as follows:

- `simlab1/sim.m`: driver script that runs all five deliverables and records summary notes
- `simlab1/sim1.m`: Deliverable 1 batch least squares implementation
- `simlab1/sim2.m`: Deliverable 2 batch least squares implementation with landmark sensing
- `simlab1/sim3.m`: Deliverable 3 odometry-only Kalman filter implementation
- `simlab1/sim4.m`: Deliverable 4 Kalman filter implementation with landmark sensing
- `simlab1/sim5.m`: Deliverable 5 back-and-forth batch least squares implementation
- `simlab1/problem_setup_orig.m`: provided starter setup file

Disclosure of Assistance

Generative AI was used as a supplementary tool to help interpret the assignment prompt and assist with editing portions of this report. The MATLAB implementations, simulation runs, figure generation, and final interpretation of the results were produced, verified, and submitted by me, Matthew Lepis.